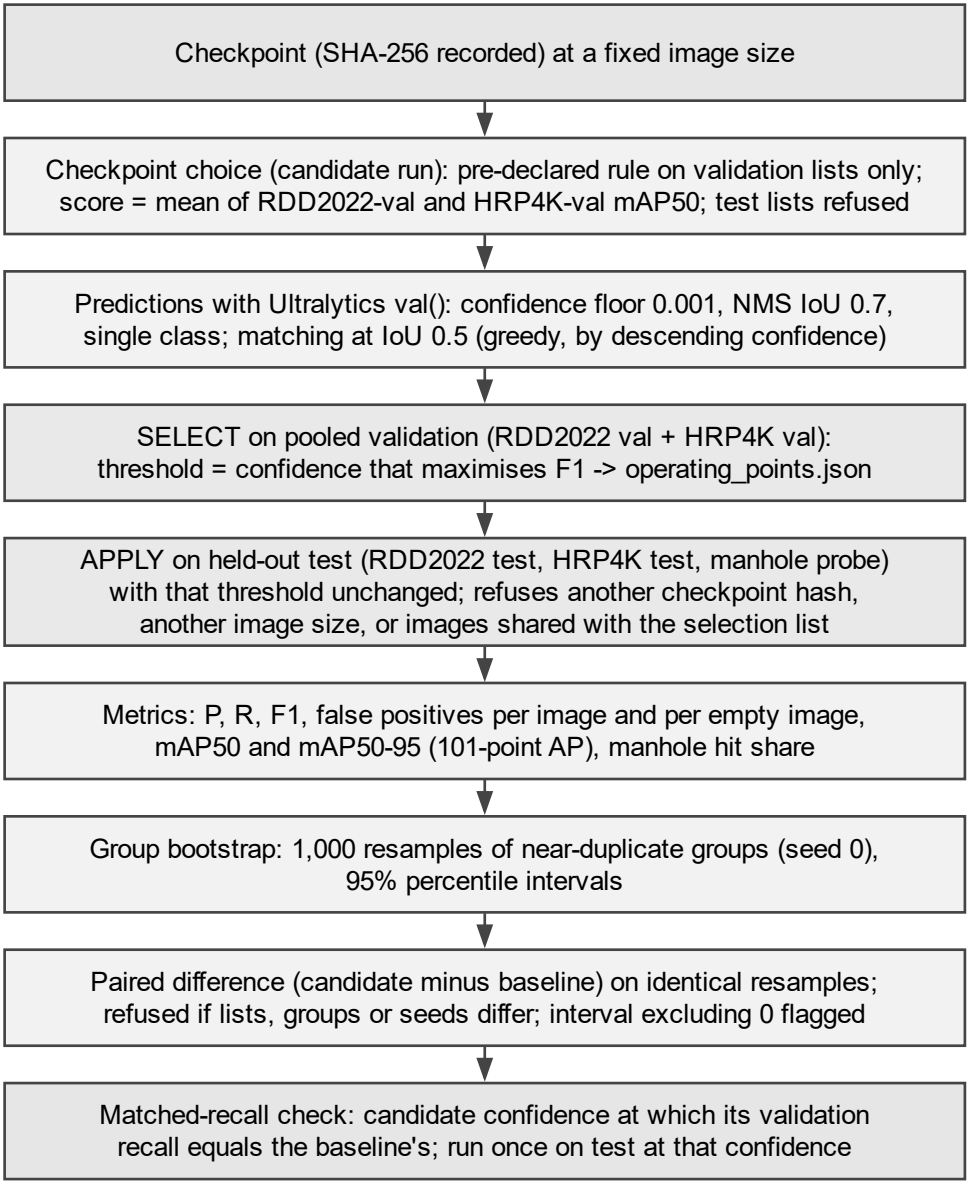
Checkpoint (SHA-256 recorded) at a fixed image size
Checkpoint choice (candidate run): pre-declared rule on validation lists only; score = mean of RDD2022-val and HRP4K-val mAP50; test lists refused
Predictions with Ultralytics val(): confidence floor 0.001, NMS IoU 0.7, single class; matching at IoU 0.5 (greedy, by descending confidence)
SELECT on pooled validation (RDD2022 val + HRP4K val): threshold = confidence that maximises F1 -> operating_points.json
APPLY on held-out test (RDD2022 test, HRP4K test, manhole probe) with that threshold unchanged; refuses another checkpoint hash, another image size, or images shared with the selection list
Metrics: P, R, F1, false positives per image and per empty image, mAP50 and mAP50-95 (101-point AP), manhole hit share
Group bootstrap: 1,000 resamples of near-duplicate groups (seed 0), 95% percentile intervals
Paired difference (candidate minus baseline) on identical resamples; refused if lists, groups or seeds differ; interval excluding 0 flagged
Matched-recall check: candidate confidence at which its validation recall equals the baseline's; run once on test at that confidence